



Continuous safety-critical control of Euler–Lagrange systems subject to multiple obstacles and velocity constraints[☆]

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ABSTRACT

This paper studies the safety-critical control problem for Euler–Lagrange (EL) systems subject to multiple ball obstacles and velocity constraints. A key strategy is to exploit the underlying cascade structure of EL systems to design a new safety-critical controller featuring an inner–outer-loop structure. In particular, the outer-loop control law is developed based on quadratic programming (QP) to avoid ball obstacles and generate velocity reference signals fulfilling the velocity limitation. Taking full advantage of the energy conservation property, a nonlinear velocity-tracking control law is designed to form the inner loop. One major difficulty is caused by the possible non-Lipschitz continuity of the standard QP algorithm when there are multiple constraints. To solve this problem, we propose a new feasible-set reshaping technique such that the refined QP algorithm with the reshaped feasible set admits a Lipschitz continuity property. Additionally, inspired by small-gain analysis, we construct a max-type Lyapunov-like function to integrate the safety constraints and the velocity-tracking error, and prove the achievement of the safety-critical control objective. The effectiveness of the proposed design is validated through numerical simulations and experiments on a 2-link planar manipulator.

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1. Introduction

1.1. Background and literature review

Ensuring safety while accomplishing complex tasks is a fundamental requirement for various engineering control systems, including those in vehicles, robotics, and industrial processes. With the rapid advancements in computing, communication, and system integration technologies, optimization-based controllers – now capable of real-time implementation – have been widely adopted to address diverse specifications in complex control systems. Notable applications include motion planning (Arkin, 1998; Choset et al., 2005; Latombe, 1991; Lynch & Park, 2017), coordination of robotic systems (Bullo et al., 2009; Mesbahi & Egerstedt, 2010; Ren & Beard, 2008; Slotine & Siciliano, 1991), and human–robot interaction (Celi et al., 2019). In these contexts, safety remains a critical and essential consideration.

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Recently, the notions of barrier functions and control barrier functions (CBFs) have attracted considerable attention from the controls and robotics community due to its capability of mathematically defining safety constraints and constructing (optimization-based) controllers to resolve the safety constraints online. In particular, Ngo et al. (2005) introduced barrier functions to integrator backstepping to address state constraints; Wieland and Allgower (2007) proposed for the first time the notion of CBFs that takes into account the effect of control inputs; Tee et al. (2009) proposed a novel asymmetrical barrier Lyapunov function for nonlinear systems in strict feedback form with an output constraint; Ames et al. (2014, 2017) proposed the notion of zeroing barrier functions, which only assumes an increasing property when the system state is outside the safe set, whereas allows weaker constraints for the system state inside the safe set; Marley et al. (2024), Xiao and Belta (2021) proposed the notion of high-order CBF, to address the case in which the barrier function depends on the control input with a high relative degree. These results have demonstrated their effectiveness in solving various safety-critical control problems for high-order nonlinear systems.

For systems with uncertain dynamics and disturbances, Xu et al. (2014) studied the robustness of CBFs, and Jankovic (2018) introduces the notion of robust CBFs. To simultaneously incorporate multiple safety constraints, Glotfelter et al. (2021), Singletary et al. (2022a) introduced nonsmooth barrier functions, and Molnar and Ames (2023) proposed an algorithmic way to create a

single continuously differentiable CBF that captures multiple state constraints.

The substantial relationship between control Lyapunov functions, as introduced in Artstein (1983), and CBFs has paved the way for the development of a multi-objective control theory (Ames et al., 2017). Specifically, for control-affine systems, CBFs can be utilized to impose linear inequality constraints on admissible control inputs to ensure safety. This naturally leads to quadratic programming (QP) formulations when quadratic objective functions are employed to incorporate primary control objectives. Typical applications can be found in mobile robots (Wang et al., 2017; Wilson et al., 2020), robotic grasping (Shaw Cortez et al., 2021; Singletary et al., 2022a), autonomous driving (Ames et al., 2017), bipedal walking (Xiong & Ames, 2021), exoskeletons (Wu et al., 2019), and human-machine interaction systems (Singletary et al., 2019). It is also worth noting that QP, as a powerful optimization tool, has also been used to design controllers for the real-time integration of various specifications not explicitly described by barrier functions (Escande et al., 2014; Mellinger & Kumar, 2011; Nakamura, 1990).

The Euler-Lagrange (EL) equation describes the motion of a wide range of physical systems. Developing a systematic approach to ensuring the safety-critical control of EL systems is highly desirable. With industrial robotic manipulators as a typical example, safety fundamentally requires that every point on the manipulator's body maintains a sufficient distance from the unsafe region. This implies that each single point in the unsafe region in the workspace often results in multiple forbidden postures, leading to multiple obstacle points in the configuration space (joint space); see Spong and Vidyasagar (2008).

One approach is to design upper-level safety mechanisms in advance to constrain the postures of EL systems within predetermined safe regions (Dietrich et al., 2012; Lozano-Perez, 1981; Youakim et al., 2017). However, such methods may restrict real-time performance in cluttered environments. On the other hand, artificial potential field methods can integrate environmental sensing feedback directly into feedback control loop but may fail to fully account for the effects of the nonlinear uncertain dynamics (Khatib, 1986).

Recently, the QP algorithm and CBFs have shown great promise in addressing safety-critical control problems for EL systems. In Singletary et al. (2022b), energy-based CBFs were introduced to extend purely kinematic CBFs by leveraging the inherent energy conservation property of EL systems. Molnar et al. (2022) proposed a two-step approach: first, a safe velocity is generated using a CBF, and then a tracking controller is designed to ensure that the actual velocity follows the safe velocity. The design naturally results in an inner-outer-loop structure. The study in Shaw Cortez and Dimarogonas (2022) focuses on the scenario where the states of the EL system are constrained within generalized axis-aligned bounding boxes and proposed a sampled-data controller and a correct-by-design algorithm to construct multiple, non-conflicting CBFs.

In a related effort, the conference paper Liu et al. (2023a) introduced the basic concept and some preliminary results on the safety-critical control of EL systems subject to multiple position constraints. However, it did not provide proofs regarding key properties such as feasibility (whether the QP problem admits at least one solution), robust safety, and Lipschitz continuity of the controller. Additionally, the conference paper Liu et al. (2023b) explored a completely different scenario, focusing on collision avoidance of the end effector of a robotic manipulator in the workspace.

1.2. The contribution of this paper

This paper takes a step toward addressing the safety-critical control of EL systems subject to multiple obstacles and velocity constraints. In particular, the obstacles are modeled as spherical regions (balls) that encapsulate unsafe areas in the configuration space. Directly applying existing designs to this problem presents considerable challenges. It remains unclear how to effectively exploit the inherent energy conservation property of EL systems when employing the existing methods developed to handle high-order dynamics and/or multiple constraints.

We propose a novel class of safety-critical controllers with an inner-outer-loop structure to tackle the aforementioned issues. Specifically, the outer-loop QP-based control law focuses on the kinematics part of the EL system, treating the generalized velocity as the virtual control input, and addresses position constraints imposed by the obstacles; the inner-loop control law is dedicated to velocity tracking control of the dynamics part of the EL system, while also accounting for velocity constraints.

In our inner-outer loop design, the output of the outer-loop QP-based control law serves as the velocity reference signal for the inner loop, and its Lipschitz continuity is essential for analyzing the convergence of the velocity-tracking error. However, a standard QP-based control law may not ensure the desired Lipschitz continuity when multiple constraints are present. Specifically, in such cases, the well-known sufficient condition for Lipschitz continuity of solutions to QP problems, as established by Hager (1979), may not hold in general. While the issue of non-Lipschitz behavior in QP-based algorithms is widely recognized and significant efforts have been made to construct continuous QP-based controllers (see, e.g., Morris et al., 2015; Shaw Cortez et al., 2021), to the best of our knowledge, the challenge of designing a continuous safety-critical controller for EL systems under multiple constraints remains unresolved. Furthermore, existing literature lacks results that provide a unified analysis of safety-critical control under multiple constraints and velocity-tracking control, while fully accounting for the energy conservation properties of EL systems.

Our contribution is highlighted by the following:

- We propose a new feasible-set reshaping technique, which projects the original safety constraints onto a predefined positive basis to satisfy a full-row-rank condition, and enables a refined QP-based control law to handle multiple obstacles and velocity constraints while ensuring Lipschitz continuity.
- We introduce a novel max-type Lyapunov-like function, which integrates barrier functions representing safety constraints and the velocity-tracking error, and guarantees the achievement of the safety-critical control objective while fully accounting for the inherent energy conservation property of the EL system.

Notations

Most of the notations in this paper follow the convention of nonlinear control and can be found in Khalil (2002). In this paper, $\|\cdot\|$ represents the Euclidean norm for real vectors and the induced 2-norm for real matrices. For a real vector x , we use $[x]_i$ to represent the i th element. For real column vectors $x_1, \dots, x_m$, the notation $[x_1; \dots; x_m]$ represents the column-stacked vector $[x_1^T, \dots, x_m^T]^T$. A continuous function $\alpha: \mathbb{R} \rightarrow \mathbb{R}$ is said to be of class $\mathcal{K}^e$ if it is strictly increasing and $\alpha(0) = 0$. The matrix inequality $A < B$ for square matrices A and B means that the matrix $B - A$ is positive definite. For a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable at $x \in \mathbb{R}^n$, its gradient at $x \in \mathbb{R}^n$ is denoted as $\nabla f(x)$. The generalized directional derivative of a locally Lipschitz

function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at $x \in \mathbb{R}^n$ with a vector $v \in \mathbb{R}^n$ is defined by

$$f^\circ(x, v) = \limsup_{y \rightarrow x, \lambda \downarrow 0} \frac{f(y + \lambda v) - f(y)}{\lambda}. \quad (1)$$

The generalized gradient of a locally Lipschitz function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at $x \in \mathbb{R}^n$ is defined by

$$\partial f(x) = \{\epsilon \in \mathbb{R}^n : f^\circ(x, v) \geq \epsilon^T v, \forall v \in \mathbb{R}^n\}. \quad (2)$$

For a vector-valued function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$, we use $f_{ci} : \mathbb{R}^m \rightarrow \mathbb{R}$, $i \in \{1, \dots, n\}$ to represent the i th component of f . The symbol conv represents the convex hull. The sum of two sets A_1 and A_2 is defined as $A_1 + A_2 = \{a_1 + a_2 : a_1 \in A_1, a_2 \in A_2\}$.

2. Problem formulation

Consider the model of an n -DOF EL system as follows:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + N(q, \dot{q}) = u, \quad (3)$$

where $q \in \mathbb{R}^n$ is the vector of generalized coordinates (positions), $M(q) \in \mathbb{R}^{n \times n}$ is the inertia matrix, $C(q, \dot{q})\dot{q}$ represents the Coriolis and centrifugal forces with $C(q, \dot{q}) \in \mathbb{R}^{n \times n}$, $N(q, \dot{q}) \in \mathbb{R}^n$ includes generalized resistance and gravity terms, and $u \in \mathbb{R}^n$ represents the vector of generalized forces. The study in this paper does not rely on known system dynamics, but instead assumes the inherent properties of EL systems; see [Craig \(2005\)](#), [Murray et al. \(1994\)](#).

Assumption 1. The system (3) admits the following properties:

- Along the trajectories of the system (3), $\dot{M} - 2C$ is a skew-symmetric matrix, that is,

$$(\dot{M} - 2C) + (\dot{M} - 2C)^T = 0; \quad (4)$$

- $M(q)$ is an invertible, symmetric matrix for any $q \in \mathbb{R}^n$ and there exist positive constants μ_{m1} and μ_{m2} such that

$$\mu_{m1}I_{n \times n} < M(q) < \mu_{m2}I_{n \times n} \quad (5)$$

for all $q \in \mathbb{R}^n$.

In our problem setting, u is considered as the control input. Given $v_c : \mathbb{R}_+ \rightarrow \mathbb{R}^n$ as the velocity command signal, our objective is to design a controller for (3) to satisfy the imposed constraints while minimally deviating from the given velocity command signal v_c . Specifically, the safety constraints on the generalized coordinates q are defined by spherical obstacles. It is desired that the following condition holds:

$$h_i(q(t)) \geq 0, \quad i = 1, \dots, N, \quad (6)$$

for all $t \geq 0$, with

$$h_i(q) = |q - q_{oi}| - d_{si}, \quad (7)$$

where $q_{oi} \in \mathbb{R}^n$ and $d_{si} > 0$ represent the positions and radii of the spherical obstacles, respectively.

Remark 1. In practice, it is normally required that every point on the body (of a robotic manipulator, for example) maintains a sufficient distance from all the points in the unsafe region in the workspace. This makes it quite difficult to explicitly represent all the safety constraints in the workspace. However, in the configuration space, with the generalized coordinate q considered as the state, it is much easier to use balls to cover the unsafe region and define barrier functions. This is the reason why the safety-critical controller is designed in the configuration space.

Moreover, the generalized velocities are required to stay within bounded ranges:

$$|\dot{q}(t)| \leq \bar{v} \quad (8)$$

for all $t \geq 0$, where $\bar{v}$ is a positive constant representing the velocity upper bound.

Without loss of generality, the velocity command signal v_c is assumed to be bounded and differentiable on the time-line.

Assumption 2. There exist positive constants $\bar{v}_c$ and v_c^d such that

$$|v_c(t)| \leq \bar{v}_c, \quad |\dot{v}_c(t)| \leq v_c^d, \quad (9)$$

for all $t \geq 0$.

Remark 2. In scenarios where a position feedback control objective needs to be achieved alongside safety requirements, a new feedback link should be established from q to v_c . In this case, building on the results of this paper, some additional Lipschitz continuity conditions on v_c with respect to q are required for the closed-loop analysis of the overall system. For an example of such an analysis, please refer to Section 5.2.

A mild distance condition is assumed on the obstacles.

Assumption 3. There exist a constant $d_a > 0$ satisfying $d_a < \min_{i=1, \dots, N} \{d_{si}\}$, a constant $d_b > 0$ and a constant $d_h > 0$ satisfying $d_h < \min_{i=1, \dots, N} \{d_{si}\}$ such that for any q satisfying $h_i(q) \geq -d_h$ for all $i = 1, \dots, N$, if

$$h_j(q) \leq \frac{d_b}{2}, \quad h_k(q) \leq \frac{d_b}{2} \quad (10)$$

hold for some $j, k \in \{1, \dots, N\}$, then

$$|q_{oj} - q_{ok}| \leq d_a. \quad (11)$$

Remark 3. In order to satisfy Assumption 3, the unsafe region in the configuration space has to be covered by ball obstacles that are either intersecting or disjoint, and the positions of the intersecting ball obstacles are close enough. For this purpose, one may first divide the configuration space into uniform-sized hypercubes and then assign balls with proper radius to the hypercubes to cover the unsafe region. Assumption 3 can be satisfied by choosing the hypercubes small enough and the balls large enough and adding balls according to the unsafe region. An example for safety-critical control of a 2-link planar manipulator is given in Section 5.

Remark 4. The EL system model (3) exhibits a cascade structure, which motivates the application of well-established constructive nonlinear control principle (see, e.g., [Sepulchre et al. \(1997\)](#)) to address the safety-critical control problem.

Based on this idea, we first treat $\dot{q}$ as the virtual control input for the kinematics part of the EL system and design a QP-based virtual control law that computes the velocity command v^* to fulfill the position constraints. Next, we design a velocity-tracking control law to tackle the dynamics part of the EL system, ensuring that the actual velocity $\dot{q}$ tracks v^* . The Lipschitz continuity of the QP-based virtual control law is essential for analyzing the convergence of the velocity-tracking error. However, a standard QP-based control law may fail to ensure the desired Lipschitz continuity when multiple constraints are present. This presents the first major technical challenge in this paper.

In the presence of the velocity tracking error

$$\tilde{v} = \dot{q} - v^*, \quad (12)$$

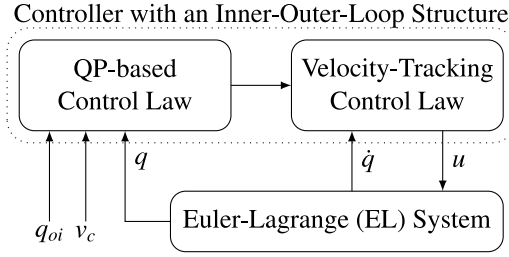


Fig. 1. The structure of the proposed safety-critical control system.

the kinematics part of the EL system takes the form

$$\dot{q} = v^* + \tilde{v}. \quad (13)$$

and the safety property should be carefully analyzed by fully taking into account the influence of $\tilde{v}$. On the other hand, the tracking control performance is also influenced by the safety-critical control part, which means the interaction between the outer-loop safety-critical control law and the inner-loop tracking control law. This presents the second major technical challenge in this paper.

3. Design of a safety-critical controller with an inner-outer-loop structure

This section presents the design procedure and the main result in this paper. The block diagram of the proposed safety-critical controller is presented in Fig. 1. The technical challenges outlined in Remark 4 are addressed through two key innovations: (a) our novel feasible-set reshaping technique, which ensures a Lipschitz continuous virtual control law, and (b) our new max-type Lyapunov-like function, which guarantees simultaneous satisfaction of the constraints on both the generalized coordinates and velocities.

3.1. QP-based control law to handle position constraints

For the kinematics part given by (13), with v^* considered as the control input, we first try using QP to integrate the safety constraints and the primary velocity command:

$$v^* = \operatorname{argmin}_{v_o^* \in \mathbb{R}^n} |v_o^* - v_c|^2 \quad (14)$$

$$\text{s.t. } -\nabla h_i(q)v_o^* \leq \alpha_p(h_i(q)) - v_r, \quad i = 1, \dots, N, \quad (15)$$

where α_p is of class $\mathcal{K}^e$, constant $v_r > 0$ is used as a robust term to handle the influence of the velocity-tracking error $\tilde{v}$, and according to the definition in (7),

$$\nabla h_i(q) = \frac{(q - q_{oi})^T}{|q - q_{oi}|} \quad \forall q \neq q_{oi}. \quad (16)$$

Remark 5. Standard results such as Ames et al. (2017) can be readily used to guarantee the feasibility of the QP problem (14)–(15) and analyze the safety property of the closed-loop system composed of (13) and (14)–(15) when $|\tilde{v}| \leq v_r$. In the case of $|\tilde{v}| \leq v_r$, condition (15) implies that along the trajectories of (13), the time-derivative of each $h_i(q)$, denoted by $\dot{h}_i(q, v)$, satisfies

$$\begin{aligned} \dot{h}_i(q, v) &= \nabla h_i(q)(v^* + \tilde{v}) \\ &\geq -\alpha_p(h_i(q)) + v_r - |\nabla h_i(q)\tilde{v}| \\ &\geq -\alpha_p(h_i(q)), \end{aligned} \quad (17)$$

which, under specific initial conditions, guarantees the satisfaction of the position constraints.

However, a normal QP-based control law may not be able to ensure the Lipschitz continuity of the solution v^* when there are more than one constraint (i.e., $N \geq 2$; see Hager (1979) and Theorem 3 in the appendix). This means that the QP algorithm (14)–(15) may not be readily used as an outer-loop control law. Moreover, the velocity constraint (8) cannot be represented in an affine form in general, and thus cannot be directly included as a constraint of the QP problem (14)–(15).

We contribute a refined feasible-set reshaping technique to solve this problem. In particular, the original safety constraints are projected on an appropriately chosen positive basis such that the resulting constraints feature a constant constraint matrix which satisfies the full-row-rank condition required by Hager (1979) for Lipschitz continuity of solutions to constrained convex optimization problems. Moreover, with the positive basis, we can represent the velocity constraint in an affine form.

The resulting QP algorithm with the reshaped feasible set is given by

$$v^* = \operatorname{argmin}_{v_o^* \in \mathbb{R}^n} |v_o^* - v_c|^2 \quad (18)$$

$$\text{s.t. } l_k^T v_o^* \leq c_0 \alpha_c(r(q, l_k)) - v_r, \quad k = 1, \dots, m \quad (19)$$

$$l_k^T v_o^* \leq c_0(\bar{v} - v_r), \quad k = 1, \dots, m \quad (20)$$

where α_c is chosen to be of class $\mathcal{K}^e$, c_0 is a positive constant less than 1 and

$$r(q, l_k) = \min_{i=1, \dots, N} \{h_i(q) + \phi(\nabla h_i(q)l_k)\}, \quad (21)$$

is called a distance-regularization function, with $\phi : [-1, 1] \rightarrow \mathbb{R}_+$ being an appropriately chosen Lipschitz continuous, non-negative, increasing function, $l_1, \dots, l_m \in \mathbb{R}^n$ are unit vectors forming a positive basis¹ such that any unit vector $l_o \in \mathbb{R}^n$ can be represented as a positive combination of the elements of $\{l_1, \dots, l_m : l_i^T l_o \geq c_0\}$, whose cardinality is not less than n . The existence of such a positive basis is guaranteed by Lemma 1 in the appendix.

The operation of $r(q, l_k)$ projects all the $\nabla h_i(q)$ for $i = 1, \dots, N$ to the positive base l_k and takes their minimum. This together with α_c of class $\mathcal{K}^e$ functions guarantees that any v_o^* satisfying the new constraints is within the original feasible set defined by (15). The function ϕ is used to enlarge the reshaped feasible set without violating the original safety constraints.

The advantage of the feasible-set reshaping technique lies in that the constraints (19)–(20), equivalently represented by

$$l_k^T v_o^* \leq \min\{c_0 \alpha_c(r(q, l_k)) - v_r, c_0(\bar{v} - v_r)\}, \quad k = 1, \dots, m, \quad (22)$$

features the constraint matrix $[l_1, \dots, l_m]^T$, any n rows of which can be guaranteed to be linearly independent (which is called the “full-row-rank condition” in this paper) by appropriately choosing the positive basis $\{l_1, \dots, l_m\}$. As proved by Hager (1979), the full-row-rank condition is sufficient to guarantee the Lipschitz continuity of the solution to the QP problem (also see Theorem 3 in the appendix of this paper).

We employ a numerical example to show the possible non-Lipschitz continuity of the solution to a QP problem and the basic idea of our feasible-set reshaping technique.

Example 1. Consider the kinematics part given by (13) with $n = 2$. Given velocity command signal $v_c = [0; 5]$ and two disk-type obstacles with the same radius $d_{si} = 0.096$ but centered at $q_{o1} = [0.1; 0]$, $q_{o2} = [-0.1; 0]$, with v^* considered as the virtual

¹ See B for the notions of positive basis.

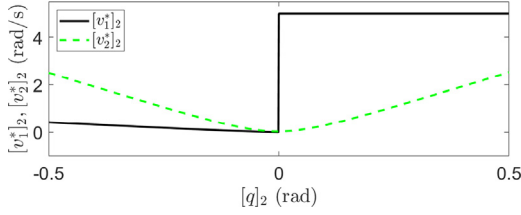


Fig. 2. The maps from $[q]_2$ to $[v^*]_2$ of different QP-based control laws.

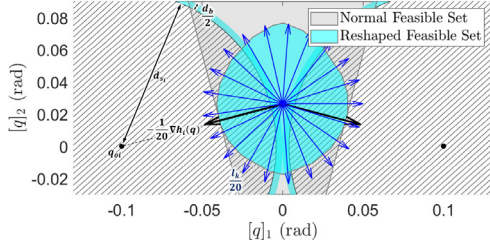


Fig. 3. An illustration of the feasible-set reshaping technique.

control input, we first design a control law based on the original QP (14)–(15) as

$$v_1^* = \operatorname{argmin}_{v_0^* \in \mathbb{R}^n} |v_0^* - v_c|^2 \quad (23)$$

$$\text{s.t. } -\nabla h_i(q)v_0^* \leq h_i(q) - v_r, \quad i = 1, 2 \quad (24)$$

with h_i defined by (7), and $v_r = 0.004$.

For comparison, we also design a control law based on the refined QP (18)–(19) but neglect the velocity constraint (20) as

$$v_2^* = \operatorname{argmin}_{v_0^* \in \mathbb{R}^n} |v_0^* - v_c|^2 \quad (25)$$

$$\text{s.t. } l_k^T v_0^* \leq c_0 \alpha_c(r(q, l_k)) - v_r, \quad k = 1, \dots, m \quad (26)$$

with $m = 21$, $l_k = [\cos(2(k-1)\pi/21); \sin(2(k-1)\pi/21)]$ for $k = 1, \dots, m$, $c_0 = 0.9556$, $\alpha(s) = 6s$ for $s \in \mathbb{R}$, $v_r = 0.004$, and

$$r(q, l_k) = \min_{i=1,2} \{h_i(q) + \max\{0.0023 \nabla h_i(q) l_k + 0.0015, 0\}\}. \quad (27)$$

Due to the space limitation, Fig. 2 only shows the map from $[q]_2$ to $[v^*]_2$. However, it can still be observed that the normal QP-based control law results in a non-Lipschitz map, while the refined QP-based control law admits a Lipschitz map.

Fig. 3 shows the q_1 – q_2 plane and v_1 – v_2 plane simultaneously for an intuitive illustration of the feasible set reshaping technique in one single figure. Note that some of the vectors are scaled by 1/20 to more effectively convey the basic idea.

If the refined QP problem (18)–(20) admits a unique solution, then we use $\varphi : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ to represent the resulting map from q and v_c to v^* , i.e.,

$$v^* = \varphi(q, v_c). \quad (28)$$

Denote

$$\underline{d}_s = \min_{i=1,\dots,N} \{d_{si}\}, \quad \bar{d}_s = \max_{i=1,\dots,N} \{d_{si}\}. \quad (29)$$

Proposition 1 shows the validity of the refined QP-based control law in dealing with multiple constraints while ensuring Lipschitz continuity of φ .

Proposition 1. Consider the kinematics part (13) of the EL system (3), and set v^* as the solution to the QP problem (18)–(20). Suppose that Assumptions 2 and 3 are satisfied. Under the condition that

$$\sqrt{2}\underline{d}_s > d_a, \quad (30)$$

$$\bar{v} > 2v_r, \quad (31)$$

$$\frac{2\underline{d}_s^2 - d_a^2}{2(\bar{d}_s + d_b/2)^2} > 1 - \frac{(\bar{v} - 2v_r)^2}{2(\bar{v} - v_r)^2} \quad (32)$$

with $\bar{v}$ defined right after (8), d_a and d_b defined in Assumption 3, and v_r being a positive constant, one can find a Lipschitz continuous, nonnegative, increasing function $\phi : [-1, 1] \rightarrow \mathbb{R}_+$, a constant $0 < c_0 < 1$, a Lipschitz function $\alpha_c \in \mathcal{K}^e$ whose inverse function is also Lipschitz, and a constant $\Delta_h > 0$ such that if

$$h_i(q) \geq -\Delta_h, \quad i = 1, \dots, N, \quad (33)$$

then

(a) the QP problem (18)–(20) admits a unique solution;

(b) the position trajectories satisfy

$$\nabla h_i(q)\dot{q} \geq -\alpha_p(h_i(q)) + v_r - |\tilde{v}|, \quad i = 1, \dots, N \quad (34)$$

with $\alpha_p(s) = \max\{\alpha_c(s), c_0\alpha_c(s)\}$, and the velocity reference signal v^* satisfies

$$|v^*| \leq \bar{v} - v_r; \quad (35)$$

(c) the resulting function φ in (28) generated by solving the QP problem (18)–(20) satisfies

$$|\varphi(q_1, v_{c1}) - \varphi(q_2, v_{c2})| \leq L(|q_1 - q_2| + |v_{c1} - v_{c2}|) \quad (36)$$

for all q_1, q_2 satisfying (33) with q replaced by q_1, q_2 , and $v_{c1}, v_{c2} \in \mathbb{R}^n$, with constant $L > 0$.

Proof. Condition (30)–(32) guarantees the existence of a $c_0 \in (0, 1)$ satisfying

$$c_0 > \sqrt{c^*} + \sqrt{c^* + \frac{v_r}{\bar{v} - v_r}}, \quad (37)$$

with

$$c^* = \frac{(2\underline{d}_s + d_b)^2 - 4\underline{d}_s^2 + 2d_a^2}{2(\bar{d}_s + d_b)^2}. \quad (38)$$

Then, we have

$$c_0(\bar{v} - v_r) > \frac{v_r}{\sqrt{c^* + v_r/(\bar{v} - v_r)} - \sqrt{c^*}}, \quad (39)$$

which means the existence of an $\alpha_c \in \mathcal{K}^e$ satisfying

$$c_0(\bar{v} - v_r) \geq c_0\alpha_c\left(\frac{d_b}{2}\right) - v_r > \frac{v_r - c_0\alpha_c(-d_h)}{\sqrt{c^* + \frac{v_r}{\bar{v} - v_r}} - \sqrt{c^*}}, \quad (40)$$

with d_h defined in Assumption 3. Moreover, with c_0 fixed, one can always find a ϕ satisfying

$$\begin{aligned} \phi(s) &> \max_{p \in [0, d_h]} \left\{ \max_{K \in \left\{ c_0(\bar{v} - v_r), \frac{v_r - c_0\alpha_c(-p)}{\sqrt{c^* + \frac{v_r}{\bar{v} - v_r}} - \sqrt{c^*}} \right\}} \left\{ \alpha_c^{-1} \left(\frac{K(s + c_0)}{c_0} \right) \right. \right. \\ &\quad \left. \left. + \frac{-K \left(\sqrt{c^* + \frac{v_r}{\bar{v} - v_r}} - \sqrt{c^*} \right) + v_r}{c_0} \right) \right\} + p \right\}, \quad s \in (-c_0, 1], \end{aligned} \quad (41)$$

$$\phi(s) = 0, \quad s \in [-1, -c_0], \quad (42)$$

With c_0 , α_c and ϕ chosen to satisfy (37)–(38) and (40)–(42), we prove the three properties of the proposition one-by-one.

(a) Feasibility of the QP Problem

Under Assumption 3, with conditions (30)–(32), (37)–(38) and (40)–(42) satisfied, we choose $\Delta_h \in (0, d_h]$, $K_v > 0$ and $c_M \in (0, 1)$ satisfying

$$\frac{2(\underline{d}_s - \Delta_h)^2 - d_a^2}{2(\bar{d}_s + d_b/2)^2} > c_M > 1 - \frac{(\bar{v} - 2v_r)^2}{2(\bar{v} - v_r)^2}, \quad (43)$$

$$c_0 > \sqrt{\frac{1-c_M}{2}} + \sqrt{\frac{1-c_M}{2} + \frac{v_r}{\bar{v}-v_r}}, \quad (44)$$

$$c_0(\bar{v}-v_r) \geq c_0\alpha_c\left(\frac{d_b}{2}\right) - v_r \geq K_v \geq \frac{-c_0\alpha_c(-\Delta_h) + v_r}{c_0 - \sqrt{2-2c_M}}, \quad (45)$$

$$c_0 - \sqrt{2-2c_M} > \sqrt{c^* + \frac{v_r}{\bar{v}-v_r}} - \sqrt{c^*}, \quad (46)$$

$$\begin{aligned} \phi(s) &> \max_{K \in \left\{c_0(\bar{v}-v_r), \frac{v_r - c_0\alpha_c(-\Delta_h)}{\sqrt{c^* + \frac{v_r}{\bar{v}-v_r}} - \sqrt{c^*}}\right\}} \left\{ \alpha_c^{-1}\left(\frac{K(s+c_0)}{c_0}\right) \right. \\ &\quad \left. + \frac{-K\left(\sqrt{c^* + \frac{v_r}{\bar{v}-v_r}} - \sqrt{c^*}\right) + v_r}{c_0} \right\} + \Delta_h, \quad s \in (-c_0, 1]. \end{aligned} \quad (47)$$

In accordance with Assumption 3, using condition (33) and $\Delta_h \in (0, d_h]$, the definition of the function r in (21) guarantees the existence of sets K_1 and K_2 satisfying

$$K_1 \cup K_2 = \{1, \dots, m\}, \quad K_1 \cap K_2 = \emptyset \quad (48)$$

such that

$$r(q, l_k) \geq d_b/2 \text{ for } k \in K_1 \text{ and } r(q, l_k) < d_b/2 \text{ for } k \in K_2. \quad (49)$$

Then, we discuss the following two cases separately.

- $k \in K_1$. In this case, due to (45), for any $v_s \in \mathbb{R}^n$ satisfying $|v_s| \leq K_v$, we have

$$l_k^T v_s \leq K_v \leq c_0\alpha_c(d_b/2) - v_r \leq c_0\alpha_c(r(q, l_k)) - v_r \quad (50)$$

for all $k \in K_1$.

- $k \in K_2$. Using (41)–(42) and (43)–(47), for $s \in [-1, 1]$, we have

$$\phi(s) \geq \max \left\{ \alpha_c^{-1} \left(\frac{K_v(s + \sqrt{2-2c_M}) + v_r}{c_0} \right) + \Delta_h, 0 \right\}. \quad (51)$$

In this case, from the definition of $h_i(q)$ in (7) and the function r in (21), under conditions (33) and (51), there exists a $J \subseteq \{1, \dots, N\}$ such that for any $i \in J$,

$$-\Delta_h \leq |q_{oi} - q| - d_{si} < d_b/2, \quad (52)$$

and for any $k \in K_2$, there exists an $i^* \in J$ such that

$$r(q, l_k) = |q_{oi^*} - q| - d_{si^*} + \phi(\nabla h_{i^*}(q)l_k) \geq -\Delta_h. \quad (53)$$

Under Assumption 3, with $\Delta_h \in (0, d_h]$, using (7), (29), (30), (33) and (43), for any $i, i' \in J$, we have

$$|q_{oi} - q_{oi'}| \in [0, d_a], \quad (54)$$

$$\begin{aligned} \nabla h_i(q) \nabla h_{i'}^T(q) &= \frac{|q_{oi'} - q|^2 + |q_{oi} - q|^2 - |q_{oi} - q_{oi'}|^2}{2|q_{oi} - q| |q_{oi'} - q|} \\ &\geq \frac{2(d_s - \Delta_h)^2 - d_a^2}{2(d_s + \frac{d_b}{2})^2} \geq c_M. \end{aligned} \quad (55)$$

Then, using the Cosine Theorem, for any $i, i' \in J$, we can prove that

$$|\nabla h_i(q) - \nabla h_{i'}(q)| \leq \sqrt{2-2c_M}. \quad (56)$$

Now, for all $i \in J$, we show that $v^* = K_v \nabla h_i(q)$ with K_v defined by (45) satisfies (19) for $k \in K_2$. Under conditions (42), (51)–(53) and (56), with q satisfying (33), we have

$$\begin{aligned} K_v \nabla h_i(q)l_k &= K_v(\nabla h_i(q) - \nabla h_{i^*}(q) + \nabla h_{i^*}(q))l_k \\ &\leq K_v \sqrt{2-2c_M} - K_v \nabla h_{i^*}(q)l_k \end{aligned}$$

$$\begin{aligned} &\leq c_0\alpha_c(\phi(\nabla h_i(q)l_k) - \Delta_h) - v_r \\ &\leq c_0\alpha_c(r(q, l_k)) - v_r \end{aligned} \quad (57)$$

holds for any $i \in J$ and $k \in K_2$.

Thus, the QP problem (18)–(20) is always feasible.

A QP problem admits a unique solution as long as it is feasible (see, e.g., Bertsekas (1999)).

(b) Satisfaction of the Position and Velocity Constraints

The positive basis $\{l_1, \dots, l_m\}$ is chosen such that for each $i = 1, \dots, N$, there exist p elements of $\{l_1, \dots, l_m\}$, say $l_{i1}, \dots, l_{ip}$, and constants $\rho_{i1}, \dots, \rho_{ip} \geq 0$, with $p \geq n$, such that

$$-\nabla h_i^T(q) = \sum_{j=1}^p \rho_{ij} l_{ij}, \quad -\nabla h_i(q)l_{ij} \geq c_0 \quad (58)$$

for $j = 1, \dots, p$, which directly implies

$$1 \leq \sum_{j=1}^p \rho_{ij} \leq \frac{1}{c_0}. \quad (59)$$

With (58) satisfied, we also have

$$-\nabla h_i(q)v^* = \sum_{j=1}^p \rho_{ij} l_{ij}^T v^*, \quad i = 1, \dots, N. \quad (60)$$

Recall the definition of the function r in (21). With the satisfaction of conditions (19), (42) and (58), for any l_{ij} of $l_{i1}, \dots, l_{ip}$, we have

$$\begin{aligned} l_{ij}^T v^* &\leq c_0\alpha_c(r(q, l_{ij})) - v_r \\ &= c_0\alpha_c\left(\min_{w=1, \dots, N} \{h_w(q) + \phi(\nabla h_w(q)l_{ij})\}\right) - v_r \\ &\leq c_0\alpha_c(h_i(q)) - v_r. \end{aligned} \quad (61)$$

Then, (19), (59), (60) and (61) together imply

$$\begin{aligned} -\nabla h_i(q)v^* &= \sum_{j=1}^p \rho_{ij} l_{ij}^T v^* \leq \sum_{j=1}^p \rho_{ij} c_0\alpha_c(r(q, l_{ij})) - \sum_{j=1}^p \rho_{ij} v_r \\ &\leq \sum_{j=1}^p \rho_{ij} c_0\alpha_c(h_i(q)) - v_r. \end{aligned} \quad (62)$$

From (58) and (62), we have that

$$-\nabla h_i(q)v^* \leq \begin{cases} \alpha_c(h_i(q)) - v_r, & \text{if } h_i(q) \geq 0, \\ c_0\alpha_c(h_i(q)) - v_r, & \text{if } h_i(q) < 0. \end{cases} \quad (63)$$

Thus, we have

$$-\nabla h_i(q)v^* \leq \alpha_p(h_i(q)) - v_r \quad (64)$$

with $\alpha_p(s) = \max\{c_0\alpha_c(s), \alpha_c(s)\}$ for $s \in \mathbb{R}$. It can be directly checked that $\alpha_p \in \mathcal{K}^e$. Then, using (7), (33) and (64), we have

$$\nabla h_i(q)v^* \geq -\alpha_p(h_i(q)) + v_r. \quad (65)$$

With each $\nabla h_i(q)$ being a unit vector, using (13) and (65), we can prove (34).

We rewrite any solution v^* of the QP problem in the form of $v^* = |v^*|l_0$ with l_0 being a unit vector. Then, the selection of the positive basis $\{l_1, \dots, l_m\}$ guarantees the existence of at least one element $l_i \in \{l_1, \dots, l_m\}$ satisfying $l_i^T l_0 \geq c_0$ such that

$$k_x l_i^T l_0 \leq c_0(\bar{v} - v_r), \quad (66)$$

which means

$$|v^*| = k_x \leq \frac{c_0(\bar{v} - v_r)}{l_i^T l_0} \leq \frac{c_0(\bar{v} - v_r)}{c_0} = \bar{v} - v_r. \quad (67)$$

(c) Lipschitz Continuity of the QP-based Control Law

Note that the constraints (19)–(20) can be equivalently represented by (22), which features the constant constraint matrix $[l_1, \dots, l_m]^T$, any n rows of which are guaranteed to be linearly independent. Then, in the region of q satisfying (33), by directly using (Hager, 1979, Theorem 3.1) (also see Theorem 3 in the appendix of this paper), we can prove the Lipschitz continuity of the solution to the QP problem with respect to q and v_c by showing that the right-hand sides of the constraints are Lipschitz continuous with respect to q . With α_c chosen to be Lipschitz continuous, this can be achieved by showing the Lipschitz continuity of $r(q, l_k)$ with respect to q .

With (7) and (21), we rewrite

$$r(q, l_k) = \min_{i=1, \dots, N} \{r_k(q, q_{oi}, d_{si})\} \quad (68)$$

with

$$r_k(q, q_{oi}, d_{si}) = |q - q_{oi}| - d_{si} + \phi(\nabla h_i(q)l_k). \quad (69)$$

With q satisfying (33), we can prove the Lipschitz continuity of $r_k(q, q_{oi}, d_{si})$ with respect to q , and thus that of $r(q, l_k)$ with respect to q . $\square$

Remark 6. Under Assumptions 2 and 3, with conditions (30)–(32) satisfied, one can construct a Lipschitz continuous QP-based safety control law (18)–(20) by firstly appropriately choosing c_0 , α_c and ϕ according to (37)–(42) and then defining a positive basis $\{l_1, \dots, l_m\} \subset \mathbb{R}^n$ such that any unit vector $l_o \in \mathbb{R}^n$ is a positive combination of the elements of $\{l_1, \dots, l_m : l_o^T l_i \geq c_0, i = 1, \dots, m\}$ whose cardinality is not less than n . The existence of such a positive basis is guaranteed by Lemma 1 in the appendix.

Remark 7. Combining safety constraints using smooth max (or min) operators as proposed in Molnar and Ames (2023) is useful to construct continuously differentiable CBF candidate functions. However, as discussed right after Theorem 5 in that paper, the resulting continuously differentiable CBF candidate, say $h(x)$, may not be valid as the CBF condition $\sup_{u \in \mathbb{R}^n} \dot{h}(x, u) \geq -\alpha(h(x))$ may be violated in certain cases of $\nabla h(x) = 0$ and $h(x) < 0$.

3.2. Velocity-tracking control law

Given the velocity reference signal v^* generated by the outer-loop control law proposed in Section 3.1, we design a velocity-tracking control law for the dynamics part of the system (3).

With the velocity-tracking error defined in (12), consider the following velocity-tracking control law

$$u = N(q, \dot{q}) + C(q, \dot{q})v^* - k_D \ddot{v} \quad (70)$$

with constant $k_D > 0$, and a Lyapunov function candidate

$$V(q, \ddot{v}) = \frac{1}{2} \ddot{v}^T M(q) \ddot{v}. \quad (71)$$

Recall (12) and (28). We equivalently rewrite $V(q, \ddot{v})$ as

$$\begin{aligned} \ddot{V}(q, \dot{q}, v_c) &= V(q, \ddot{v}) \\ &= \frac{1}{2} (\dot{q} - \varphi(q, v_c))^T M(q) (\dot{q} - \varphi(q, v_c)). \end{aligned} \quad (72)$$

The following proposition shows that the controlled EL system admits a velocity-tracking capability.

Proposition 2. Under Assumptions 1 and 2, consider the system (3) and the control law (70). Suppose the existence of a positive constant L such that the outer-loop control law given by $v^* = \varphi(q, v_c)$ satisfies (36) for all $q_1, v_{c1}, q_2, v_{c2} \in \mathbb{R}^n$. Then,

$$\delta^T [\dot{q}; \ddot{q}; \dot{v}_c]$$

$$\leq -\frac{2k_D}{\mu_{m2}} \ddot{V}(q, \dot{q}, v_c) + 2nL\mu_{m2} \sqrt{\frac{2\ddot{V}(q, \dot{q}, v_c)}{\mu_{m1}}} (v_c^d + |\dot{q}|) \quad (73)$$

holds for any $q, \dot{q}, \ddot{q}, v_c, \dot{v}_c \in \mathbb{R}^n$, and any $\delta \in \partial \ddot{V}(q, \dot{q}, v_c)$.

Proof. Rewrite the control law $\varphi(q, v_c)$ as

$$\varphi(q, v_c) = [\varphi_{c1}(q, v_c); \dots; \varphi_{cn}(q, v_c)]. \quad (74)$$

With the existence of the positive constant L for the satisfaction of (36), using Clarke (1990, Proposition 2.1.1), one can prove

$$|\varphi_{ci}^\circ(q, v_c, v)| \leq 2L|v|, \quad i = 1, \dots, n \quad (75)$$

for all $q, v_c \in \mathbb{R}^n$ and $v \in \mathbb{R}^{2n}$, where $\varphi_{ci}^\circ(q, v_c, v)$ represents the directional derivative of $\varphi_{ci}(q, v_c)$. Then, according to the definition of generalized gradients in (2),

$$2L \geq |\epsilon|, \quad \forall \epsilon \in \partial \varphi_{ci}(q, v_c), \quad i \in \{1, \dots, n\} \quad (76)$$

for $q, v_c \in \mathbb{R}^n$. We rewrite $\ddot{V}$ defined in (72) as

$$\ddot{V}(q, \dot{q}, v_c) = \frac{1}{2} V_1(q, \dot{q}, v_c) - V_2(q, \dot{q}, v_c) + \frac{1}{2} V_3(q, \dot{q}, v_c) \quad (77)$$

with

$$V_1(q, \dot{q}, v_c) = \dot{q}^T M(q) \dot{q}, \quad (78)$$

$$V_2(q, \dot{q}, v_c) = \dot{q}^T M(q) \varphi(q, v_c), \quad (79)$$

$$V_3(q, \dot{q}, v_c) = \varphi(q, v_c)^T M(q) \varphi(q, v_c). \quad (80)$$

Using the sum rule of generalized gradients (see Clarke (1990, Proposition 2.3.3)) yields

$$\partial \ddot{V}(q, \dot{q}, v_c) \subset \frac{1}{2} \partial V_1(q, \dot{q}, v_c) - \partial V_2(q, \dot{q}, v_c) + \frac{1}{2} \partial V_3(q, \dot{q}, v_c). \quad (81)$$

Under Assumptions 1 and 2, with (28), (70), (72) and (76)–(81), by using the product rule of generalized gradients in Clarke (1990, Proposition 2.3.13), we have

$$\begin{aligned} \delta^T [\dot{q}; \ddot{q}; \dot{v}_c] &\leq \dot{q}^T M(q) \ddot{q} + \frac{1}{2} \dot{q}^T \dot{M}(q) \dot{q} - \dot{q}^T \dot{M}(q) \varphi(q, v_c) - \varphi^T(q, v_c) M(q) \ddot{q} \\ &\quad + 2nL\mu_{m2} |\dot{q} - \varphi(q, v_c)| (|\dot{q}| + |\dot{v}_c|) + \frac{1}{2} \varphi^T(q, v_c) \dot{M}(q) \varphi(q, v_c) \\ &\leq -k_D |\dot{q} - \varphi(q, v_c)|^2 + 2nL\mu_{m2} |\dot{q} - \varphi(q, v_c)| (|\dot{q}| + |\dot{v}_c|) \\ &\leq -\frac{2k_D}{\mu_{m2}} \ddot{V}(q, \dot{q}, v_c) + 2nL\mu_{m2} \sqrt{\frac{2\ddot{V}(q, \dot{q}, v_c)}{\mu_{m1}}} (v_c^d + |\dot{q}|), \end{aligned} \quad (82)$$

for all $\delta \in \partial \ddot{V}(q, \dot{q}, v_c)$. This proves the satisfaction of property (73) for any $q, \dot{q}, v_c \in \mathbb{R}^n$, and any $\delta \in \partial \ddot{V}(q, \dot{q}, v_c)$. $\square$

Various velocity tracking control laws have been proposed in the literature. In our paper, the velocity tracking control law is designed as part of a systematic solution to address the safety control problem for EL systems subject to multiple constraints. Other designs may further enhance performance and reduce the complexity of the control architecture (Molnar et al., 2022).

3.3. Synthesis of the inner–outer-loop controller

In this section, we show the validity of the safety controller composed of the outer-loop safety control law (18)–(20) in Section 3.1 and the inner-loop velocity-tracking control law (70) in Section 3.2. In particular, we construct a new max-type Lyapunov-like function to integrate the safety constraints and the velocity-tracking error, and guarantee the achievement of the safety-critical control objective.

Theorem 1. Under Assumptions 1, 2 and 3, the closed-loop system composed of the EL system (3) and the controller consisting of (18)–(20) and (70) admits the following properties:

- Under the condition (30)–(32), with $\bar{v}$ defined right after (8), d_a and d_b defined in Assumption 3, and v_r being a positive constant, there exist a Lipschitz continuous, nonnegative, increasing function $\phi : [-1, 1] \rightarrow \mathbb{R}_+$, a constant $0 < c_0 < 1$, a Lipschitz function $\alpha_c \in \mathcal{K}^e$ whose inverse function is also Lipschitz, and a positive constant Δ_h such that the map $v^* = \varphi(q, v_c)$ representing the solution of (18)–(20) satisfies

$$|\varphi(q_1, v_{c1}) - \varphi(q_2, v_{c2})| \leq L(|q_1 - q_2| + |v_{c1} - v_{c2}|) \quad (83)$$

for all q_1, q_2 satisfying (33) with q replaced by q_1, q_2 and all $v_{c1}, v_{c2} \in \mathbb{R}^n$, with L being a positive constant;

- In addition, with k_D chosen such that

$$k_D > 2nL \frac{\mu_{m2}^3(v_c^d + \bar{v})}{\mu_{m1}^2 v_r}, \quad (84)$$

there exist a Lipschitz $\mathcal{K}^e$ function α_w and a positive constant d_w such that

$$\delta^T[\dot{q}; \ddot{q}; \dot{v}_c] \leq -\alpha_w(W(q, \dot{q}, v_c)), \quad \forall \delta \in \partial W(q, \dot{q}, v_c), \quad (85)$$

holds for all $q, \dot{q}, v_c$ satisfying $W(q, \dot{q}, v_c) \leq d_w$, where

$$W(q, \dot{q}, v_c) = \max\{W_1(q, \dot{q}, v_c), \dots, W_{N+1}(q, \dot{q}, v_c)\}, \quad (86)$$

with

$$W_i(q, \dot{q}, v_c) = -h_i(q), \quad i = 1, \dots, N, \quad (87)$$

$$W_{N+1}(q, \dot{q}, v_c) = -\frac{nL\mu_{m1}\mu_{m2}^2(v_c^d + \bar{v} - v_r)v_r}{\mu_{m1}k_D - 2nL\mu_{m2}^2} + \frac{1}{2}(\dot{q} - \varphi(q, v_c))^T M(q)(\dot{q} - \varphi(q, v_c)). \quad (88)$$

Proof. Condition (83) is directly guaranteed by Proposition 1. Now, we prove the satisfaction of (85)–(88).

Define

$$J_w(q, \dot{q}, v_c) = \{i \in \{1, \dots, N+1\} : W_i(q, \dot{q}, v_c) = W(q, \dot{q}, v_c)\}. \quad (89)$$

Using the pointwise maxima rule and sum rule of generalized gradients (see, e.g., Clarke (1990, Proposition 2.3.12)), we have

$$\partial W(q, \dot{q}, v_c) \subset \text{conv}\{\partial W_i(q, \dot{q}, v_c), i \in J_w(q, \dot{q}, v_c)\}. \quad (90)$$

Under condition (90), using the definition of convex hull in Grünbaum et al. (1967), there exist appropriately chosen constants λ_i , for $i \in J_w(q, \dot{q}, v_c)$ satisfying

$$\sum_{i \in J_w(q, \dot{q}, v_c)} \lambda_i = 1, \quad \min_{i \in J_w(q, \dot{q}, v_c)} \lambda_i \geq 0, \quad (91)$$

and appropriately chosen vectors $\delta_i \in \partial W_i(q, \dot{q}, v_c)$, $i \in J_w(q, \dot{q}, v_c)$ such that for any $\delta \in \partial W(q, \dot{q}, v_c)$,

$$\delta = \sum_{i \in J_w(q, \dot{q}, v_c)} \lambda_i \delta_i. \quad (92)$$

The condition (84) leads to

$$\frac{\mu_{m1}^2 v_r^2}{2\mu_{m2}} - \frac{nL\mu_{m1}\mu_{m2}^2(v_c^d + \bar{v} - v_r)v_r}{\mu_{m1}k_D - 2nL\mu_{m2}^2} > 0. \quad (93)$$

Thus, we can choose d_w satisfying

$$d_w \leq \min\left\{\Delta_h, \frac{\mu_{m1}^2 v_r^2}{2\mu_{m2}} - \frac{nL\mu_{m1}\mu_{m2}^2(v_c^d + \bar{v} - v_r)v_r}{\mu_{m1}k_D - 2nL\mu_{m2}^2}\right\}. \quad (94)$$

Under the condition that

$$W(q, \dot{q}, v_c) \leq d_w, \quad (95)$$

with Assumption 1 satisfied, using (86), (88) and (95), we have

$$|\dot{q} - \varphi(q, v_c)| \leq \sqrt{\frac{\mu_{m1}}{\mu_{m2}}} v_r, \quad (96)$$

$$h_i(q) \geq -\Delta_h, \quad i = 1, \dots, N. \quad (97)$$

Consider the following two cases:

- $N+1 \notin J_w(q, \dot{q}, v_c)$

In this case, with the satisfaction of Assumption 1, from Proposition 1, under conditions (89)–(97) satisfied, there exists a Lipschitz, class $\mathcal{K}^e$ function α_{w1} such that

$$\begin{aligned} \delta^T[\dot{q}; \ddot{q}; \dot{v}_c] &= \sum_{i \in J_w(q, \dot{q}, v_c)} -\lambda_i \nabla h_i(q) \dot{q} \\ &\leq \sum_{i \in J_w(q, \dot{q}, v_c)} \lambda_i [\alpha_p(h_i(q)) - v_r + |\dot{q} - \varphi(q, v_c)|] \\ &\leq \sum_{i \in J_w(q, \dot{q}, v_c)} \lambda_i \alpha_p(h_i(q)) \\ &= \sum_{i \in J_w(q, \dot{q}, v_c)} -\lambda_i \alpha_{w1}(W_i(q, \dot{q}, v_c)) \\ &= -\alpha_{w1}(W(q, \dot{q}, v_c)) \end{aligned} \quad (98)$$

for any $\delta \in \partial W(q, \dot{q}, v_c)$

- $N+1 \in J_w(q, \dot{q}, v_c)$

By using techniques similar to those in (98), we have for $\delta_i \in \partial W_i(q, \dot{q}, v_c)$, $i \in J_w(q, \dot{q}, v_c) \setminus \{N+1\}$,

$$\delta_i^T[\dot{q}; \ddot{q}; \dot{v}_c] \leq -\alpha_{w1}(W(q, \dot{q}, v_c)). \quad (99)$$

With Assumptions 1 and 2 satisfied, by using Propositions 1 and 2, there exists a Lipschitz, class $\mathcal{K}^e$ function α_{w2} such that

$$\begin{aligned} \delta_{N+1}^T[\dot{q}; \ddot{q}; \dot{v}_c] &\leq -\frac{2k_D}{\mu_{m2}} \check{V}(q, \dot{q}, v_c) + 2nL\mu_{m2} \sqrt{\frac{2\check{V}(q, \dot{q}, v_c)}{\mu_{m1}}} (v_c^d + |\dot{q}|) \\ &\leq -\frac{2k_D}{\mu_{m2}} \left(W_{N+1}(q, \dot{q}, v_c) + \frac{nL\mu_{m1}\mu_{m2}^2(v_c^d + \bar{v} - v_r)v_r}{\mu_{m1}k_D - 2nL\mu_{m2}^2} \right) \\ &\quad + 2nL\mu_{m2} \sqrt{\frac{2\check{V}(q, \dot{q}, v_c)}{\mu_{m1}}} \left(v_c^d + \bar{v} - v_r + \sqrt{\frac{2\check{V}(q, \dot{q}, v_c)}{\mu_{m1}}} \right) \\ &\leq -\frac{2k_D}{\mu_{m2}} \left(W_{N+1}(q, \dot{q}, v_c) + \frac{nL\mu_{m1}\mu_{m2}^2(v_c^d + \bar{v} - v_r)v_r}{\mu_{m1}k_D - 2nL\mu_{m2}^2} \right) \\ &\quad + 4nL \frac{\mu_{m2}}{\mu_{m1}} \left(W_{N+1}(q, \dot{q}, v_c) + \frac{nL\mu_{m1}\mu_{m2}^2(v_c^d + \bar{v} - v_r)v_r}{\mu_{m1}k_D - 2nL\mu_{m2}^2} \right) \\ &\quad + 2nL\mu_{m2}(v_c^d + \bar{v} - v_r) \sqrt{\frac{2\check{V}(q, \dot{q}, v_c)}{\mu_{m1}}} \\ &= -\left(\frac{2k_D}{\mu_{m2}} - 4nL \frac{\mu_{m2}}{\mu_{m1}} \right) W_{N+1}(q, \dot{q}, v_c) \\ &\quad - \left(\frac{2k_D}{\mu_{m2}} - 4nL \frac{\mu_{m2}}{\mu_{m1}} \right) \frac{nL\mu_{m1}\mu_{m2}^2(v_c^d + \bar{v} - v_r)v_r}{\mu_{m1}k_D - 2nL\mu_{m2}^2} \\ &\quad + 2nL\mu_{m2}(v_c^d + \bar{v} - v_r) \sqrt{\frac{2\check{V}(q, \dot{q}, v_c)}{\mu_{m1}}} \\ &\leq -\left(\frac{2k_D}{\mu_{m2}} - 4nL \frac{\mu_{m2}}{\mu_{m1}} \right) W_{N+1}(q, \dot{q}, v_c) \\ &\quad + 2nL\mu_{m2}(v_c^d + \bar{v} - v_r) \left(\sqrt{\frac{2\check{V}(q, \dot{q}, v_c)}{\mu_{m1}}} - v_r \right) \end{aligned}$$

$$\begin{aligned} &\leq -\left(\frac{2k_D}{\mu_{m2}} - 4nL\frac{\mu_{m2}}{\mu_{m1}}\right)W_{N+1}(q, \dot{q}, v_c) \\ &\leq -\alpha_{w2}(W(q, \dot{q}, v_c)) \end{aligned} \quad (100)$$

for any $\delta_{N+1} \in \partial W_{N+1}(q, \dot{q}, v_c)$, where we have used conditions (12), (72), (83), (86)–(88), and (95)–(97) for the first inequality, used conditions (5), (35) (86)–(88), and (95)–(97) for the second inequality, used condition (88) for the third inequality, and used conditions (5), (84), (86) and (95)–(96) for the fifth inequality.

Thus, for any $\delta_i \in W_i(q, \dot{q}, v_c)$, $i \in J_w(q, \dot{q}, v_c)$, we have

$$\delta_i^T[\dot{q}; \ddot{q}; \dot{v}_c] \leq -\alpha_w(W(q, \dot{q}, v_c)) \quad (101)$$

with

$$\alpha_w(s) = \min\{\alpha_{w1}(s), \alpha_{w2}(s)\}. \quad (102)$$

From (89)–(97) and (101), for any $\delta \in \partial W(q, \dot{q}, v_c)$, we have

$$\begin{aligned} \delta^T[\dot{q}; \ddot{q}; \dot{v}_c] &= \sum_{i \in J_w(q, \dot{q}, v_c)} \lambda_i \delta_i^T[\dot{q}; \ddot{q}; \dot{v}_c] \\ &\leq -\alpha_w(W(q, \dot{q}, v_c)). \end{aligned} \quad (103)$$

This ends the proof of Theorem 1. $\square$

Remark 8. Property (85) implies

$$\dot{W}(q(t), \dot{q}(t), v_c(t)) \leq -\alpha_w(W(q(t), \dot{q}(t), v_c(t))) \quad (104)$$

for almost every $t \geq 0$, as long as $W(q(t), \dot{q}(t), v_c(t)) \leq d_w$ (see, e.g., Moreau and Valadier (1987, Theorem 2)). Also with Lakshmikantham and Leela (1969, Theorem 1.10.2), there exists a $\beta \in \mathcal{KL}$ satisfying $\beta(s, 0) = s$ for all $s \geq 0$ such that if initially $W(q(0), \dot{q}(0), v_c(0)) \leq d_w$, then

$$W(q(t), \dot{q}(t), v_c(t)) \leq \beta(\max\{W(q(0), \dot{q}(0), v_c(0)), 0\}, t) \quad (105)$$

for all $t \geq 0$. This means that

$$W(q(t), \dot{q}(t), v_c(t)) \leq 0, \quad \forall t \geq 0 \quad (106)$$

provided that $W(q(0), \dot{q}(0), v_c(0)) \leq 0$. Then, using the definition of W in (86) and (87), we can guarantee the achievement of the safety-critical control objectives (6) and (8).

Remark 9. The max-type combination (86) of the barrier functions and the Lyapunov function of different subsystems of the closed-loop system is somehow motivated by the Lyapunov-based nonlinear small-gain theorem (Jiang et al., 1996), in which max-type Lyapunov functions are constructed for interconnected systems based on Lyapunov functions of the subsystems.

The following example shows how the safety-critical controller based on the refined QP is capable of guaranteeing the safety of the controlled EL system subject to multiple obstacles.

Example 2. This example still considers the case in Example 1 but also takes into account the dynamics part of the EL system (3). Suppose that

$$M(q) = \frac{1}{100} \begin{bmatrix} 11 + 6 \cos[q]_2 & 9 + 3 \cos[q]_2 \\ 9 + 3 \cos[q]_2 & 9 \end{bmatrix}, \quad (107)$$

$$C(q, \dot{q}) = \frac{1}{100} \begin{bmatrix} -6[\dot{q}]_2 \sin[q]_2 & -3[\dot{q}]_2 \sin[q]_2 \\ 3[\dot{q}]_1 \sin[q]_2 & 0 \end{bmatrix}, \quad (108)$$

$$N(q, \dot{q}) = \begin{bmatrix} 1.96 \cos[q]_1 + 2.94 \cos([q]_1 + [q]_2) \\ 2.94 \cos([q]_1 + [q]_2) \end{bmatrix}, \quad (109)$$

and the control input $u \in \mathbb{R}^2$.

We propose two safety control laws of the form in (70), where $k_D = 40$ and $\tilde{v} = v_i^* - \dot{q}$ for $i = 1, 2$. Here, v_i^* for $i = 1, 2$

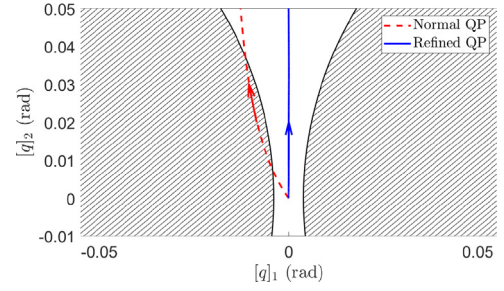


Fig. 4. Position trajectories of the EL system with the different controllers. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

are the solutions to the original QP (23)–(24) and the refined QP (25)–(26), respectively.

Fig. 4 shows the position trajectories of the controlled EL system with the different controllers, depicted as a red dashed line and a blue solid line, respectively. With the same velocity-tracking control law, collision happens with the safety-critical controller based on the normal QP (14)–(15), while safety is guaranteed by the one based on the refined QP (18)–(20).

4. A refined safety-critical controller with reduced computational complexity

The QP-based control law in the previous section takes into account all obstacles, the number of which could be very large. The computational complexity required by safety control would be significantly increased as the number of obstacles becomes large. This section shows that the obstacles with positions far away enough from the plant state do not influence the position constraints much, and proposes a refined control law to reduce the computational complexity.

Recall the function ϕ and the unit vectors l_k for $k = 1, \dots, m$, defined right after (21). We replace the function r in (21) with

$$r_f(q, l_k, d_f) = \min \left\{ \min_{i \in J_N(q, d_f)} \{h_i(q) + \phi(\nabla h_i(q)l_k)\}, d_f \right\}, \quad (110)$$

where

$$J_N(q, d_f) = \{i \in \{1, \dots, N\} : h_i(q) \leq d_f\} \quad (111)$$

with d_f being a positive constant to be determined later. Then, for $k = 1, \dots, m$, we define

$$r_k(q, d_f) = \begin{cases} r_f(q, l_k, d_f), & \text{if } J_N(q, d_f) \neq \emptyset, \\ d_f, & \text{if } J_N(q, d_f) = \emptyset, \end{cases} \quad (112)$$

and consider the following refined QP-based control law:

$$v^* = \underset{v_o^* \in \mathbb{R}^n}{\operatorname{argmin}} |v_o^* - v_c|^2 \quad (113)$$

$$\text{s.t. } l_k^T v_o^* \leq c_o \alpha_c(r_k(q, d_f)) - v_r, \quad k = 1, \dots, m \quad (114)$$

where α_c is a class $\mathcal{K}^e$ function.

With only the obstacles the distances of which are smaller than or equal to an appropriately chosen positive constant d_f taken into account, the resulting feasible velocities are naturally restricted within the desired range. This means that the boundedness constraint on the velocity need not be given explicitly in the QP problem; see property (b) of Proposition 3.

Proposition 3. Consider the kinematics part (13) of the EL system (3) with $v^* \in \mathbb{R}^n$ being the solution of the QP problem (113)–(114) and $\tilde{v} \in \mathbb{R}^n$ represents the generalized velocity-tracking

error. Suppose that Assumptions 2 and 3 are satisfied. Under conditions (30)–(32), where $\bar{v}$ is defined right after (8), d_a and d_b are defined in Assumption 3, and v_r is a positive constant, there exist a Lipschitz continuous, nonnegative, increasing function $\phi : [-1, 1] \rightarrow \mathbb{R}_+$, a positive constant $0 < c_0 < 1$, a positive constant d_f , a Lipschitz function $\alpha_c \in \mathcal{K}^e$ whose inverse function is also Lipschitz, and a constant $\Delta_h > 0$ such that if

$$h_i(q) \geq -\Delta_h, \quad i = 1, \dots, N, \quad (115)$$

then

- (a) the QP problem (113)–(114) admits a unique solution;
 (b) the position trajectories satisfy

$$\nabla h_i(q)\dot{q} \geq -\alpha_p(h_i(q)) + v_r - |\bar{v}|, \quad i = 1, \dots, N \quad (116)$$

with

$$\alpha_p(s) = \max\{\alpha_c(s), c_0\alpha_c(s)\}, \quad (117)$$

and the velocity reference signal v^* satisfies

$$|v^*| \leq \bar{v} - v_r; \quad (118)$$

- (c) the map φ in (28) generated by solving the QP problem (113)–(114) satisfies

$$|\varphi(q_1, v_{c1}) - \varphi(q_2, v_{c2})| \leq L(|q_1 - q_2| + |v_{c1} - v_{c2}|) \quad (119)$$

for all q_1, q_2 satisfying (115) with q replaced by q_1, q_2 , and $v_{c1}, v_{c2} \in \mathbb{R}^n$, where L is a positive constant.

Proof (Sketch of Proof). The parameters c_0, α_c and ϕ can be chosen similarly as for Proposition 1 such that (37)–(42) are satisfied. Then, there exists a positive constant d_f satisfying

$$\alpha_c^{-1} \left(\frac{-c_0\alpha_c(-d_h) + \left(1 + \sqrt{c^* + \frac{v_r}{\bar{v}-v_r}} - \sqrt{c^*}\right) v_r}{c_0 \left(\sqrt{c^* + \frac{v_r}{\bar{v}-v_r}} - \sqrt{c^*}\right)} \right) < d_f \leq \alpha_c^{-1} \left(\bar{v} + \left(\frac{1}{c_0} - 1\right)v_r \right), \quad (120)$$

with c^* defined in (38) and d_h defined in Assumption 3.

Then, the three properties can be proved one-by-one.

(a) Feasibility of the QP Problem

Conditions (30)–(32), (37)–(38), (40)–(42) and (120) guarantee that there exist constants $K_v > 0$, $c_M \in (0, 1)$ and $\Delta_h \in (0, d_h]$ satisfying

$$\begin{aligned} c_0(\bar{v} - v_r) &\geq c_0\alpha_c \left(\max \left\{ d_f, \frac{d_b}{2} \right\} \right) - v_r \\ &\geq c_0\alpha_c \left(\min \left\{ d_f, \frac{d_b}{2} \right\} \right) - v_r \geq K_v \geq \frac{-c_0\alpha_c(-\Delta_h) + v_r}{c_0 - \sqrt{2 - 2c_M}}, \end{aligned} \quad (121)$$

$$\phi(s) \geq \alpha_c^{-1} \left(\frac{K_v(s + \sqrt{2 - 2c_M}) + v_r}{c_0} \right) + \Delta_h \quad \forall s \in [-1, 1]. \quad (122)$$

According to conditions (112), (121)–(122), using the proof similar to Proposition 1, there exists a $v_x \in \mathbb{R}^n$ satisfying

$$|v_x| = K_v, \quad (123)$$

such that for all $k = 1, \dots, m$, if $J_N(q, d_f) \neq \emptyset$, then

$$l_k^T v_x \leq c_0\alpha_c(r_f(q, l_k, d_f)) - v_r, \quad (124)$$

and if $J_N(q, d_f) = \emptyset$, then

$$l_k^T v_x \leq c_0\alpha_c(d_f) - v_r \leq c_0(\bar{v} - v_r). \quad (125)$$

Thus, the QP problem (113)–(114) is always feasible.

A QP problem admits a unique solution as long as it is feasible (see, e.g., Bertsekas (1999)).

(b) Satisfaction of the Position and Velocity Constraints

Suppose that $v^* = k_x l_0$, where k_x is a positive constant and $l_0 \in \mathbb{R}^n$ is some unit vector. Then, due to (111)–(114) and the definition of positive basis right after (21), there exists at least one positive basis $l_k \in \{l_1, \dots, l_m\}$ satisfying $l_k^T l_0 \geq c_0$ such that

$$k_x l_k^T l_0 \leq c_0\alpha_c(d_f) - v_r. \quad (126)$$

From (126), we have

$$|v^*| = k_x \leq \frac{c_0\alpha_c(d_f) - v_r}{l_k^T l_0} \leq \alpha_c(d_f) - \frac{v_r}{c_0} = \bar{v} - v_r. \quad (127)$$

This obviously implies that the solution to the QP problem is bounded. Moreover, for any $i \in \{1, \dots, N\} \setminus J_N(q, d_f)$, by using (111), (117), (127) and $0 < c_0 < 1$, we have

$$\begin{aligned} -\nabla h_i(q)v^* &\leq \alpha_c(d_f) - \frac{v_r}{c_0} \leq \alpha_c(h_i(q)) - v_r \\ &= \alpha_p(h_i(q)) - v_r. \end{aligned} \quad (128)$$

Then, also using (13), we have

$$\nabla h_i(q)\dot{q} \geq -\alpha_p(h_i(q)) + v_r - |\bar{v}| \quad (129)$$

for $i \in \{1, \dots, N\} \setminus J_N(q, d_f)$. Applying the same techniques as for the proof of Proposition 1, we have

$$-\nabla h_i(q)v^* \leq \alpha_p(|q_{oi} - q| - d_{si}) - v_r \quad (130)$$

for $i \in J_N(q, d_f)$. The satisfaction of (116) is a direct consequence of (7), (12), (13), (128) and (130).

(c) Lipschitz Continuity of the Refined QP-based Control Law

From (21), (110) and (111), for any $q \in \mathbb{R}^n$, if $r_f(q, d_f, l_k) = d_f$, we have

$$r(q, l_k) \geq d_f, \quad (131)$$

and if $r_f(q, d_f, l_k) = \min_{i \in J_N(q, d_f)} \{h_i(q) + \phi(\nabla h_i(q)l_k)\}$, we have

$$r_f(q, d_f, l_k) = r(q, l_k) < d_f. \quad (132)$$

Thus, for any $q_1, q_2 \in \mathbb{R}^n$ satisfying (115), constant d_f satisfying (120) and l_k for $k = 1, \dots, m$ defined right after (21),

$$|r_f(q_1, d_f) - r_f(q_2, d_f)| \leq |r(q_1, l_k) - r(q_2, l_k)|. \quad (133)$$

Then, we can prove the existence of a constant $L > 0$ for (119) based on the reasoning for the proof of Proposition 1. $\square$

The following theorem shows the effectiveness of the refined safety-critical controller with a reduced computational complexity.

Theorem 2. Under Assumptions 1, 2 and 3, the closed-loop system composed of the EL system (3) and inner–outer-loop controller consisting of (113)–(70) and (114) admits the following properties:

- Under the condition (30)–(32), where $\bar{v}$ is defined right after (8), d_a and d_b are defined in Assumption 3, and v_r is a positive constant, there exist a Lipschitz continuous, nonnegative, increasing function $\phi : [-1, 1] \rightarrow \mathbb{R}_+$, a positive constant $0 < c_0 < 1$, a positive constant d_f and a Lipschitz function $\alpha_c \in \mathcal{K}^e$ whose inverse function is also Lipschitz and a positive constant $\Delta_h > 0$ such that the map $v^* = \varphi(q, v_c)$ representing the solution of (113)–(114) satisfies (83) for all $q_1, q_2 \in \mathbb{R}^n$ satisfying (115) with q replaced by q_1, q_2 and all $v_{c1}, v_{c2} \in \mathbb{R}^n$, with L being a positive constant;
- In addition, with k_D satisfying (84), there exists a Lipschitz $\mathcal{K}^e$ function α_w and a positive constant d_w such that (85)–(88) hold for all $q, \dot{q}, v_c \in \mathbb{R}^n$.

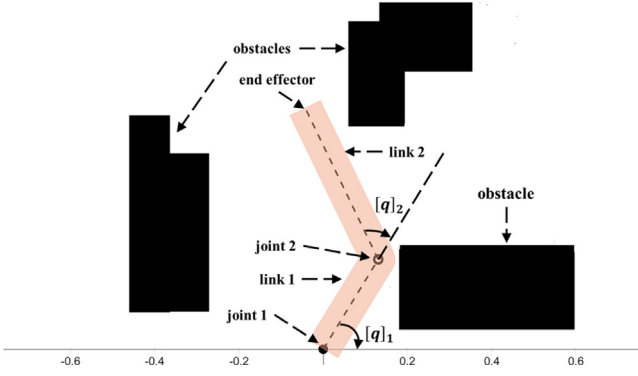


Fig. 5. A scenario of safety-critical control of a 2-link robotic manipulator.

5. Numerical and experimental validations

We consider a scenario of safety-critical control of a 2-link planar manipulator, as shown in Fig. 5. Such a scenario is ubiquitous in practice. It is required that every point on the body of the manipulator keeps far away enough from the obstacles. Moreover, with $P_2 \in \mathbb{R}^2$ and $P_e \in \mathbb{R}^2$ being the positions of joint 2 and end effector, safety also requires that $[P_2]_2 \geq 0$ and $[P_e]_2 \geq 0$. Link 1 is a uniform rod with width 0.07 m, length 0.25 m and mass 2 kg; link 2 is a uniform rod with width 0.07 m, length 0.4 m and mass 3 kg. SI units are used in this section, and the units are omitted where it does not lead to confusion.

5.1. A safety-critical control scenario and controller design

The positions q_{oi} of the ball obstacles in the joint (configuration) space are generated based on the following rule. The joint space is first divided by uniform-sized hypercubes, and balls with the same radius are assigned to the hypercubes to cover the unsafe region. In particular, the side lengths of all the hypercubes are chosen as 0.0698 rad and the radii of all the balls as 0.1396 rad.

We design two controllers whose outer-loop safety control laws are in the form of (18)–(20) and (113)–(114), respectively. For both of the controllers, following Propositions 1 and 3 and Remark 6, we choose

$$\begin{aligned} d_a &= 0.1561, & d_b &= 0.0182, & v_r &= 0.0105, & d_h &= 0.00002, \\ d_{si} &= 0.1745, & c_0 &= 0.9965, & \bar{v} &= 0.3142, \\ \alpha_c(r) &= \begin{cases} 0.9r, & \text{for } r \leq 0, \\ 34.506r, & \text{for } 0 < r \leq 0.0091, \\ 0.9(r - 0.0091) + 0.3136, & \text{for } 0.0091 < r, \end{cases} \\ \phi(r) &= \max\{-0.3362s + 0.3350, 0\}, & r &\in [-1, 1]. \end{aligned}$$

Moreover, for the controller whose outer-loop safety control law is in the form of (113)–(114), we choose

$$d_f = 0.0098.$$

Then, Assumption 3 can be verified.

5.2. Numerical simulation

The coefficients of the EL model (3) for the robotic manipulator can be readily calculated following the standard modeling approach (Craig, 2005). Regarding the 2-link planar manipulator, the coefficients are available in Mahil and Durra (2016). Moreover, we design the inner-loop velocity-tracking control law in the form of (70) with $k_D = 800.0000$.

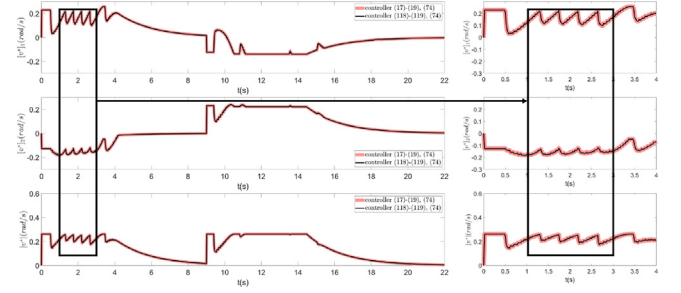


Fig. 6. Trajectories of the velocity reference signal v^* in the numerical simulation.

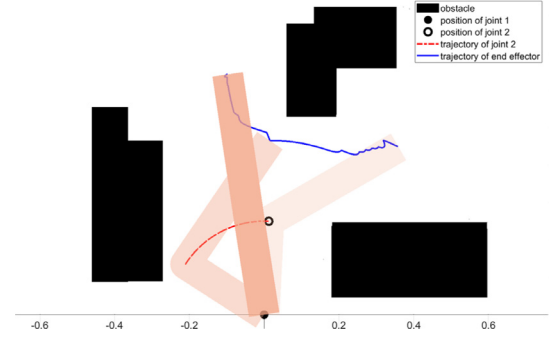


Fig. 7. The motion of the 2-link robotic manipulator in the workspace.

The primary controller is designed based on saturated proportional feedback and an appropriately chosen waypoint to keep the primary velocity command within the desired range and avoid deadlock:

$$v_{ci} = (q - q_{ri}) \max \left\{ -0.5236, -\frac{\bar{v}_c}{|q - q_{ri}|} \right\}, \quad (134)$$

with $\bar{v}_c$ being the upper bound, q_{r1} being the waypoint, and q_{r2} being the target point. In the numerical simulation, the state q approaches from the initial state $q(0) = [1.5208, -1]^T$ to the target point $q_{r2} = [1.7208, 0]^T$ through the waypoint $q_{r1} = [2.6, -1.6]^T$. Here, we set $\bar{v}_c = 0.2618$.

Note that v_c , as defined in (134), explicitly depends on q , which means that its time-derivative $\dot{v}_c$ (when defined) also depends on $\dot{q}$. This implies that the system setup in the numerical simulation differs from that considered in the main result. In this case, an intuitive solution is to consider replacing the constant v_c^d in property (9) with a function $v_c^d(|\dot{q}|)$. If the function v_c^d is sufficiently small, then the reasoning in the proof of Theorem 1 would remain valid. The validity of the proposed controller is demonstrated for the system setup through numerical simulation. However, providing a rigorous proof is nontrivial.

Fig. 6 shows the trajectories of the velocity reference signal v^* with two different safety controllers, one consisting of (18)–(20) and (70), and the other one consisting of (113)–(70) and (114). Both velocity reference signals are kept within the desired range, and coincide with each other. Figs. 7 and 8 show the motion in the workspace and in the joint space of the manipulator with controller (113)–(70) and (114), respectively. It can be observed that the motion of the manipulator keeps far away enough from the ball obstacles.

Since the velocity reference signals v^* generated by different controllers are quite close to each other, the first controller can also guarantee the safety of the manipulator. The simulation result is not given here due to space limitation.

We also test the computation times of the two controllers using different numbers of ball obstacles to represent the unsafe

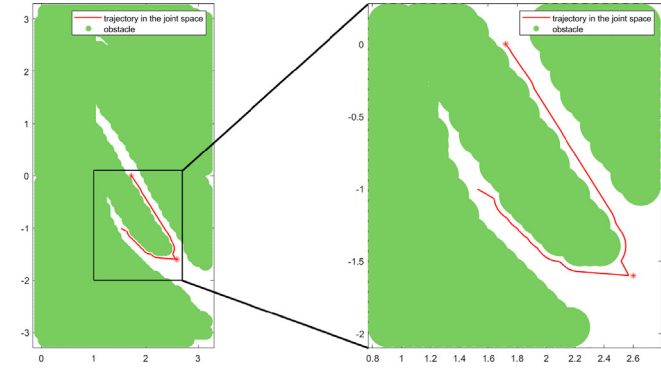


Fig. 8. Unsafe regions and the motion of the 2-link robotic manipulator in the joint space.

Table 1

Computation time of the controllers (ms).

N	Type of controller	
	(18)–(20), (70)	(113)–(114), (70)
3457	15.5	6.7
8747	38.9	6.9
13617	46.5	8.3

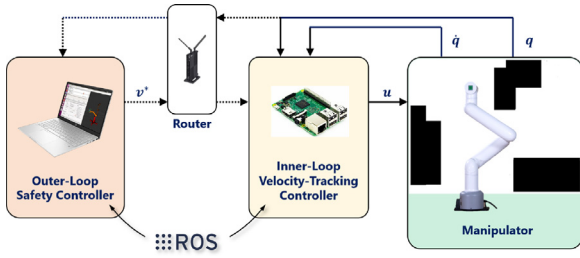


Fig. 9. The experimental set-up.

regions. Following Remark 3 and the discussions at the beginning of Section 5.1, by setting the side length of the hypercubes as 0.0698 rad, 0.0436rad and 0.0349rad, the unsafe region in the configuration space is covered by 3457, 8747 and 13617 ball obstacles, respectively. MATLAB-based numerical simulations are performed on a Windows laptop with an Intel Core i7-10510U CPU @1.80 GHz. The computation time of the two controllers per iteration are given by Table 1. It is shown that reducing the number of ball obstacles helps save computation time.

5.3. Experimental results

In this subsection, we verify our control methods based on the experimental system shown in Fig. 9. The experimental system is composed of a 6-DOF manipulator, a laptop and a wireless router. We use the second and third joints of the manipulator and lock the other four joints to verify the effectiveness of the proposed methods.

In our experiment setup, the QP-based safety-critical controllers in the form of (18)–(20) and (113)–(114) are implemented on the laptop, which communicates with the manipulator through a wireless local area network. Since the manipulator is already equipped with an inner-loop velocity-tracking controller, we only verify the outer-loop QP-based virtual control law. We perform system identification for the velocity tracking loop using Matlab System Identification Tool-box with v^* as the input and $\dot{q}$ as the output. Then, following Propositions 1 and 3 and Remark

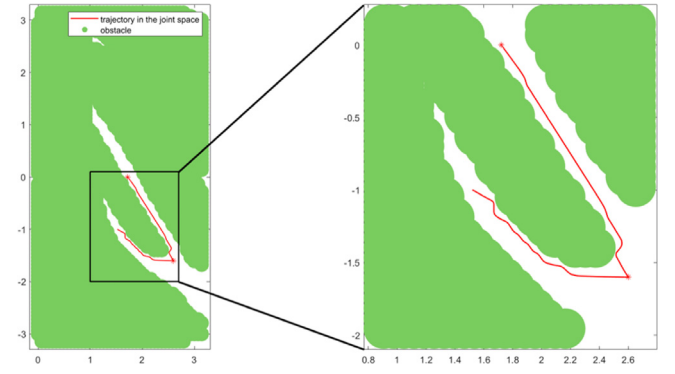


Fig. 10. Unsafe regions and the motion of the 6-link robotic manipulator in the joint space with the safety-critical controller (113)–(114).

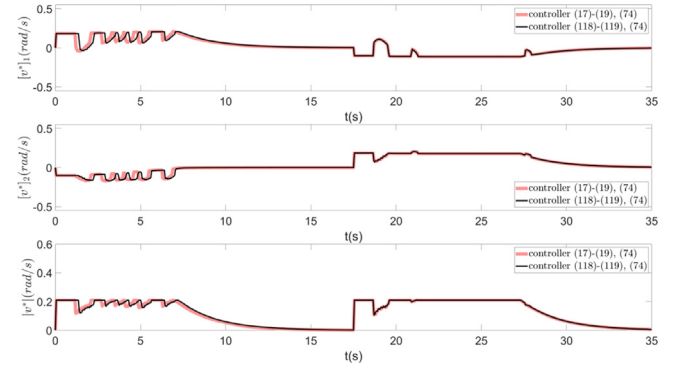


Fig. 11. Trajectories of the velocity reference signals v^* in the experiment.

6, we modify some controller parameters as follows:

$$\begin{aligned}
 v_r &= 0.0209, & \bar{v} &= 0.5585, & \bar{v}_c &= 0.2094, \\
 d_f &= 0.0091, & d_h &= 0.000002, \\
 \alpha_c(r) &= \begin{cases} 1.8r, & \text{for } r \leq 0, \\ 61.5382r, & \text{for } 0 < r \leq 0.0091, \\ 1.8(r - 0.0091) + 0.5585, & \text{for } 0.0091 < r, \end{cases} \\
 \phi(r) &= \max\{-1.4881s + 1.4828, 0\}, \quad r \in [-1, 1].
 \end{aligned}$$

Fig. 10 shows the motion of the manipulator in the joint space with controller (113)–(114). Fig. 11 shows the trajectories of the velocity reference signals v^* generated by the controllers (18)–(20) and (113)–(114), respectively. It can be observed that, with both controllers, the velocity reference signals keep within the desired range, and the manipulator keeps far away enough from the ball obstacles.

6. Conclusions

This paper has proposed a systematic solution to the safety control of EL systems subject to multiple obstacles and velocity constraints. The proposed constraint-satisfaction controller comprises a refined QP-based outer-loop control law to handle the position constraints generated by multiple obstacles and a nonlinear inner-loop control law for velocity-tracking. The feasible set of the QP-based control law is reshaped based on an appropriately chosen positive basis to overcome the fundamental challenge caused by the possibly non-Lipschitz continuity of a standard QP-based algorithm when there are multiple constraints. Inspired by the nonlinear small-gain theorem, the main result is proved by constructing a max-type Lyapunov-like function for the closed-loop system.

We believe that the result in this paper provides a solid foundation for future research on other types of EL systems that address practical engineering challenges, including handling various position constraints while considering the energy conservation property. Although external disturbances are not explicitly addressed here, the proposed methods — such as the feasible-set reshaping technique and the max-type Lyapunov-like function inspired by small-gain theory — align well with robust control principles and could be extended to handle such disturbances. The problem becomes more complex if the shapes and sizes of obstacles change over time. In such cases, if ball obstacles are used to approximate the obstacles, both their number and positions would vary with time. A potential solution would involve designing a safety-critical controller that ensures robustness against time-varying obstacles.

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Appendix A. QP and Lipschitz continuity of its solution

An optimization problem is called a QP problem if the objective function is quadratic, and the constraints are affine. A QP problem depending on external variables $x \in \mathcal{X} \subseteq \mathbb{R}^m$ and $v \in \mathcal{V} \subseteq \mathbb{R}^n$ can be expressed in the form:

$$u = \underset{u^* \in \mathbb{R}^n}{\operatorname{argmin}} u^{*T} P u^* + v^T u^* \quad (\text{A.1})$$

$$\text{s.t. } A(x)u^* \leq b(x) \quad (\text{A.2})$$

where $P \in \mathbb{R}^{n \times n}$ is a symmetric positive definite matrix, $A : \mathcal{X} \rightarrow \mathbb{R}^{N \times n}$ and $b : \mathcal{X} \rightarrow \mathbb{R}^N$. The QP problem admits a unique solution if it is feasible (i.e., at least one u^* satisfies (A.2)).

The following theorem plays a vital role in analyzing the Lipschitz continuity property of the QP problems in our paper.

Theorem 3. (Hager, 1979, Theorem 3.1) Consider the QP problem defined by (A.1) and (A.2). For each $x \in \mathcal{X}$, we represent the non-redundant active constraints² as $\tilde{A}(x)u = \tilde{b}(x)$. Suppose that the following conditions are satisfied:

- $\{A(x) : x \in \mathcal{X}\}$ and $\{b(x) : x \in \mathcal{X}\}$ are convex;
- For each $x \in \mathcal{X}$, the QP problem admits a unique solution;
- There exists a constant $\bar{A} > 0$ such that $|\tilde{A}^T(x)| < \bar{A}$ for all $x \in \mathcal{X}$;
- There exists a constant $\underline{A} > 0$ such that for each $x \in \mathcal{X}$, $|\tilde{A}^T(x)\lambda| \geq \underline{A}|\lambda|$ for all λ .

Then, with the map from x to u denoted by $\rho : \mathcal{X} \rightarrow \mathbb{R}^n$, there exist constants $L_1, L_2 > 0$ such that

$$\begin{aligned} |\rho(x_1) - \rho(x_2)| &\leq L_1(|v_1 - v_2| + |b(x_1) - b(x_2)|) \\ &\quad + L_2|A(x_1) - A(x_2)|(|v_1| + |v_2| + |b(x_1)| + |b(x_2)|) \end{aligned} \quad (\text{A.3})$$

for any $x_1, x_2 \in \mathcal{X}$ and any $v_1, v_2 \in \mathcal{V}$.

Appendix B. Positive basis

Positive bases are used for feasible-set reshaping in this paper. Relevant notions are reviewed here to make the paper self-contained. See Davis (1954) for the original development of the theory.

Definition 1. A positive combination of $a_j \in \mathbb{R}^n$ for $j = 1, \dots, r$ is a linear combination $\lambda_1 a_1 + \dots + \lambda_r a_r$ with $\lambda_j \geq 0$; it is a strictly positive combination if $\lambda_j > 0$ for $j = 1, \dots, r$.

Definition 2. A set of vectors $\{a_1, \dots, a_r\}$ is positively dependent if one of the a_i is a positive combination of the others; otherwise, the set is positively independent.

Definition 3. A set of positively independent vectors $\{a_1, \dots, a_r\}$ is a positive basis for $\mathbb{R}^n$ if each vector in $\mathbb{R}^n$ is a positive combination of $a_1, \dots, a_r$.

The following lemma is useful to construct positive bases to perform the feasible-set reshaping technique in this paper.

Lemma 1. (Wu et al., 2022, Section V.A) For specific constant c_0 satisfying $0 < c_0 < 1$, there exists a matrix $A_L = [l_1, \dots, l_{n_l}]^T$ with $l_i \in \mathbb{R}^n$ for $i = 1, \dots, n_l$ and $n_l > n$ such that

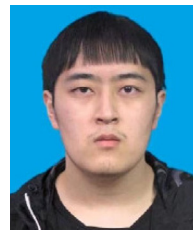
1. Each l_i is a unit vector, i.e., $|l_i| = 1$;
2. Any n vectors selected from $l_1, \dots, l_{n_l}$ are linearly independent;
3. Any unit vector $l_0 \in \mathbb{R}^n$ is a positive combination of the elements of $\{l_i : l_i^T l_0 \geq c_0, i = 1, \dots, n_l\}$, whose cardinality is not less than n_u .

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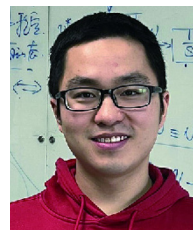
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² See Boyd and Vandenberghe (2004) for relevant notions.

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